ES114 Data Synthesis Narrative 3

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***Abstract*—The present article entails an analysis of of 8 files containing the match statistics for both women and men at the four major tennis tournaments of the year 2013. A total of eight different aspects from each dataset is intuitively compared or optimized, with the expectation of identifying the probability of an event taking place in future. The intuitions are subsequently verified by arranging and comparing the given data, utilizing conditional probability, various probability density functions, correlation coefficiet, kmeans clustering,linear regressions and various other figures, and by drawing conclusion from all of them.**

I. Overview Of The Dataset

This is a collection of 8 files containing the match statistics for both women and men at the four major tennis tournaments of the year 2013. Each file has 42 columns and a minimum of 76 rows.

Player 1 Name of Player 1 Player 2 Name of Player 2 Result Result of the match (0/1) - Referenced on Player 1 is Result = 1 if Player 1 wins (FNL.1>FNL.2) FSP.1 First Serve Percentage for player 1 (Real Number) FSW.1 First Serve Won by player 1 (Real Number) SSP.1 Second Serve Percentage for player 1 (Real Number) SSW.1 Second Serve Won by player 1 (Real Number) ACE.1 Aces won by player 1 (Numeric-Integer) DBF.1 Double Faults committed by player 1 (Numeric-Integer) WNR.1 Winners earned by player 1 (Numeric) UFE.1 Unforced Errors committed by player 1 (Numeric) BPC.1 Break Points Created by player 1 (Numeric) BPW.1 Break Points Won by player 1 (Numeric) NPA.1 Net Points Attempted by player 1 (Numeric) NPW.1 Net Points Won by player 1 (Numeric) TPW.1 Total Points Won by player 1 (Numeric) ST1.1 Set 1 result for Player 1 (Numeric-Integer) ST2.1 Set 2 Result for Player 1 (Numeric-Integer) ST3.1 Set 3 Result for Player 1 (Numeric-Integer) ST4.1 Set 4 Result for Player 1 (Numeric-Integer) ST5.1 Set 5 Result for Player 1 (Numeric-Integer) FNL.1 Final Number of Games Won by Player 1 (Numeric-Integer) FSP.2 First Serve Percentage for player 2 (Real Number) FSW.2 First Serve Won by player 2 (Real Number) SSP.2 Second Serve Percentage for player 2 (Real Number) SSW.2 Second Serve Won by player 2 (Real Number) ACE.2 Aces won by player 2 (Numeric-Integer) DBF.2 Double Faults committed by player 2 (Numeric-Integer) WNR.2 Winners earned by player 2 (Numeric) UFE.2 Unforced Errors committed by player 2 (Numeric) BPC.2 Break Points Created by player 2 (Numeric) BPW.2 Break Points Won by player 2 (Numeric) NPA.2 Net Points Attempted by player 2 (Numeric) NPW.2 Net Points Won by player 2 (Numeric) TPW.2 Total Points Won by player 2 (Numeric) ST1.2 Set 1 result for Player 2 (Numeric-Integer) ST2.2 Set 2 Result for Player 2 (Numeric-Integer) ST3.2 Set 3 Result for Player 2 (Numeric-Integer) ST4.2 Set 4 Result for Player 2 (Numeric-Integer) ST5.2 Set 5 Result for Player 2 (Numeric-Integer) FNL.2 Final Number of Games Won by Player 2 (Numeric-Integer) Round Round of the tournament at which game is played (Numeric-Integer)

Features

II. Details of Libraries And Functions

There are various libraries in python, which helps us to perform different operations on our Dataset.The Python Standard Library contains the exact syntax, semantics, and tokens of Python. It contains built-in modules that provide access to basic system functionality like I/O and some other core modules. Most of the Python Libraries are written in the C programming language. used with correspondence/brief/technote papers. The various categories of options will now be discussed. For each category, the default option is shown in bold. The user must specify an option from each category in which the default is not desired. The various categories are orthogonal to each other—changes in one will not affect the defaults in the others.By utilizing the powerful Python libraries I could perform an in-depth analysis of the given data set. The libraries provided pre-written code and functionality to perform complex tasks quickly and efficiently. After analyzing the data set, I was able to draw a well-informed conclusion based on the insights and trends discovered through the data analysis.

A. Numpy [1]

The name "Numpy" stands for "Numerical Python". It is the commonly used library. It is a popular machine learning library that supports large matrices and multi-dimensional data. It consists of in-built mathematical functions for easy computations. Even libraries like TensorFlow use Numpy internally to perform several operations on tensors.Array Interface is one of the key features of this library.

- Creation of arrays: NumPy allows you to create arrays of different dimensions and data types. You can create arrays from a list, tuple, or any other sequence using the np.array() function.

- Indexing and slicing arrays: You can access individual elements or a subset of elements from a NumPy array using indexing and slicing. You can also use Boolean indexing to select elements based on a condition.

- Mathematical operations: NumPy provides a wide range of mathematical operations such as addition, subtraction, multiplication, division, exponentiation, and logarithm. These operations can be performed on arrays or individual elements of an array.

- Broadcasting: NumPy allows you to perform operations between arrays of different shapes and sizes by automatically broadcasting the smaller array to match the shape of the larger array.

- Aggregation operations: NumPy provides many aggregation operations such as sum, mean, median, min, max, and standard deviation. These operations can be performed on an entire array or along a particular axis of an array.

- Reshaping and Transposing: NumPy allows you to reshape an array or transpose it to change its dimensions or orientation. You can use the reshape() function to change the shape of an array, and the transpose() function to flip the dimensions of an array.

B. Matplotlib [2]

Matplotlib is a cross-platform, data visualization and graphical plotting library for Python and its numerical extension NumPy. As such, it offers a viable open source alternative to MATLAB. Developers can also use matplotlib's APIs (Application Programming Interfaces) to embed plots in GUI applications.

- Creating a plot: You can create a plot using the plt.plot() function. This function takes one or more arrays of data as input and creates a line plot.

- Customizing a plot: Matplotlib provides many options for customizing a plot. You can change the color, line style, marker style, and other properties of the plot using the various functions provided by the library.

- Adding labels and titles: You can add labels to the x-axis and y-axis of a plot using the plt.xlabel() and plt.ylabel() functions, respectively. You can also add a title to the plot using the plt.title() function.

- Adding legends: You can add a legend to a plot using the plt.legend() function. This function takes a list of labels as input and creates a legend for the plot.

- Creating subplots: Matplotlib allows you to create multiple plots within a single figure using the plt.subplots() function. This function takes the number of rows and columns of subplots as input and returns a figure and an array of subplots.

- Saving a plot: You can save a plot as an image file using the plt.savefig() function. This function inputs the file name and file format and saves the plot in the specified format.

C. Pandas [3]

'Pandas' is an open source library for data analysis in Python. It provides high-level data structures and functions to quickly manipulate, analyze, and visualize complex datasets. 'Pandas' is used for various tasks such as data wrangling, cleaning, manipulation, and analysis. It can also be used for time series analysis, statistical computing, and machine learning. 'Pandas' has an intuitive API that makes it easy to work with and understand. It is also highly optimized for performance, so it can handle large datasets efficiently.

- Creating a DataFrame: A DataFrame is a two-dimensional table-like data structure in Pandas that can hold multiple types of data. You can create a DataFrame from a dictionary, a list of dictionaries, a NumPy array, or a CSV file using the pd.DataFrame() function.

- Reading and Writing Data: Pandas provides functions to read and write data from various file formats such as CSV, Excel, JSON, SQL, and HDF5. The pd.readcsv() function is commonly used to read CSV files, while the pd.tocsv() function is used to write data to a CSV file.

- Indexing and Selecting Data: You can access and select data from a DataFrame using various indexing methods such as loc[], iloc[], and []. These methods allow you to select rows and columns based on their labels or integer positions.

- Data Cleaning and Transformation: Pandas provides many functions for cleaning and transforming data such as removing missing values, replacing values, merging and joining data, and grouping data. The dropna() function is used to remove missing values from a DataFrame, while the merge() function is used to combine two DataFrames based on a common key.

- Aggregation and Statistical Analysis: Pandas provides many functions for aggregating and analyzing data such as groupby(), pivottable(), describe(), min(), max(), mean(), std(), and many others. These functions allow you to calculate summary statistics, group data by one or more columns, and perform various statistical operations on the data.

- Visualization: Pandas provides functions for creating various types of plots and charts such as line plots, scatter plots, bar charts, and histograms. These functions use the Matplotlib library to create the plots.

D. Scipy [4]

SciPy provides algorithms for optimization, integration, interpolation, eigenvalue problems, algebraic equations, differential equations, statistics, and many other classes of problems.SciPy wraps highly-optimized implementations written in low-level languages like Fortran, C, and C++. Enjoy the flexibility of Python with the speed of compiled code.SciPy's high-level syntax makes it accessible and productive for programmers from any background or experience level.

- Integration: Scipy provides functions for numerical integration, such as quad() and dblquad(), which can be used to integrate a function over single or multiple dimensions, respectively.

- Optimization: Scipy provides functions for finding the minimum or maximum of a function, such as minimize(), fmin(), and basin hopping (). These functions can be used to find the minimum or maximum of a function with or without constraints.

- Linear Algebra: Scipy provides functions for linear algebra, such as solve() and eig(), which can be used to solve linear equations and eigenvalue problems, respectively.

- Signal Processing: Scipy provides functions for signal processing, such as convolve() and correlate(), which can be used to filter and process signals.

- Fourier Transform: Scipy provides functions for Fourier transform, such as fft() and ifft(), which can convert a time-domain signal into the frequency domain and vice versa.

- Statistics: Scipy provides functions for statistical operations, such as ttest1samp() and ttestind(), which can be used to perform various statistical tests.

E. SeaBorn [5]

Seaborn is a library for making statistical graphics in Python. It builds on top of matplotlib and integrates closely with pandas data structures. Seaborn helps you explore and understand your data. Its plotting functions operate on dataframes and arrays containing whole datasets and internally perform the necessary semantic mapping and statistical aggregation to produce informative plots. Its dataset-oriented, declarative API lets you focus on what the different elements of your plots mean, rather than on the details of how to draw them.

- Distribution plots: Seaborn provides various distribution plots to visualize the distribution of a single variable or multiple variables in a dataset. Examples of these plots include histograms, kernel density plots, and rug plots.

- Categorical plots: Seaborn also provides various categorical plots for visualizing the distribution of one or more variables across different categories. Examples of these plots include bar, count, and box plots.

- Regression plots: Seaborn includes functions for creating regression plots that show the relationship between two variables. Examples of these plots include scatter plots, line plots, and residual plots.

- Heatmaps: Seaborn provides heatmap functions to create color-coded matrix visualizations of data, such as correlation matrices or confusion matrices.

- Facet grids: Seaborn includes functions for creating facet grids that allow you to plot multiple subsets of a dataset in a grid of subplots, which can be useful for exploring relationships between variables in different subsets of the data.

F. SciKit Learn [6]

Scikit-learn, also known as sklearn, is a popular open-source machine learning library for the Python programming language. It provides a range of supervised and unsupervised learning algorithms, as well as various tools for data preprocessing, model selection, and evaluation. Sklearn is widely used in the industry and academia for a variety of machine learning tasks, including classification, regression, clustering, and dimensionality reduction.

- Data preprocessing: Sklearn provides various functions for data preprocessing, including scaling, encoding, imputing missing values, and feature

selection.

- Supervised learning: Sklearn has a wide range of supervised learning algorithms, including linear regression, logistic regression, decision trees, random forests, support vector machines (SVMs), and neural networks. These algorithms can be used for classification, regression, and multi-class problems.

- Unsupervised learning: Sklearn also provides various unsupervised learning algorithms, including clustering algorithms such as k-means and hierarchical clustering, as well as dimensionality reduction techniques such as principal component analysis (PCA) and t-distributed stochastic neighbor embedding (t-SNE).

- Model selection and evaluation: Sklearn provides tools for model selection and evaluation, including cross-validation, hyperparameter tuning, and various metrics for evaluating model performance such as accuracy, precision, recall, F1 score, and AUC-ROC.

- Pipeline: Sklearn also provides a pipeline function that allows users to chain together multiple preprocessing and modeling steps into a single object, simplifying the machine learning workflow.

III. Hypotheses

Considering my observations, I aim to draw inferences among distinct categories by analyzing various graphs and tables derived from the Data Set.

A. Can we predict a player's first serve percentage based on their number of winners and unforced errors in the Wimbledon men 2013 Dataset?

B. If Player 1 commits 10 unforced errors in the match, what is the probability that they will still win the match in French Open men 2013?

C. Find the joint probability distribution of the number of games played and the total points earned by a player in the Wimbledon-women-2013 dataset?

D. What is the probability that Player 1 will win given that they have won more net points than Player 2 in AusOpen-men-2013 dataset?

E. What is the distribution of first-serve percentages for both players? What is the probability that a player with a high FSP value will also have a high FSW VALUE in US Open-men-2013?

F. Is there a correlation between a player's net points attempted and their number of winners earned in FrenchOpen-women-2013 dataset?

G. What is the probability of a player having a second serve percentage given that he/she has a high number of total points in USOpen-women-2013 dataset?

H. What is the probability that a player wins a match given that they won more points in the first set than their opponent in Aus-Women-2013 dataset?

IV. Answer to the Hypothesis

A. Can we predict a player's first serve percentage based on their number of winners and unforced errors in the Wimbledon men 2013 Dataset?

The linear regression obtained in Fig.1 can be used to predict the first-serve percentage of a player based on the number of winner points and unforced errors they make. The R-squared score of the model number of winnings and first serve percentage is 0.007813519620087583, while the R-squared score of the model unforced errors and first serve percentage(Fig.2) is 0.0014726490678289572. The respective R-squared scores show us that first serve percentage depends more on a total number of former winnings.

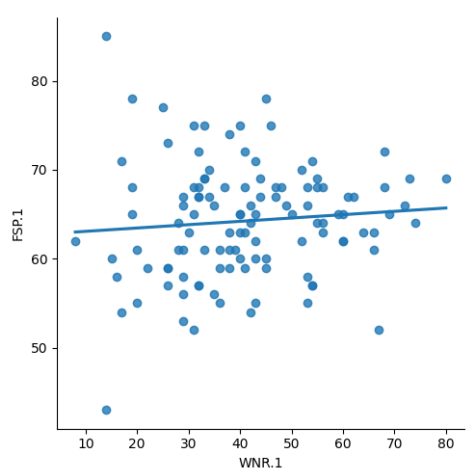

Figure 1.

Thus we can conclude that there is a slight linear relationship in both models, but a more linear relationship

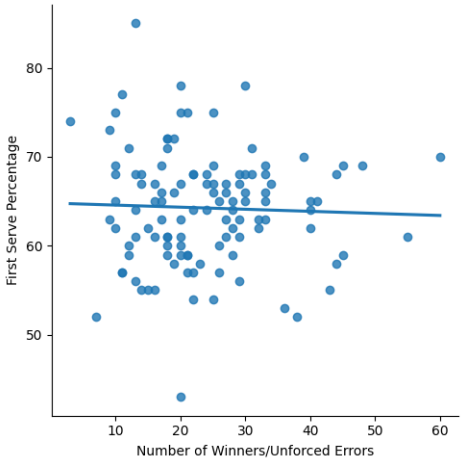

Figure 2.

exists in the total number of former winnings and first-serve percentage.

B. If Player 1 commits 10 unforced errors in the match, what is the probability that they will still win the match in French Open men 2013?

Unforced errors are mistakes that are supposedly not forced by good shots of an opponent.

Fig.3 shows a Gaussian or normal distribution of the frequency of unforced errors in the French Open Men 2013 dataset. The curve is bell-shaped, symmetric around the mean, and is centered around the mean value of the dataset. The standard deviation of the dataset indicates how spread out the data is. The plot shows that the majority of players had a moderate number of unforced errors, with fewer players having very low or very high numbers of unforced errors. The average number of unforced errors

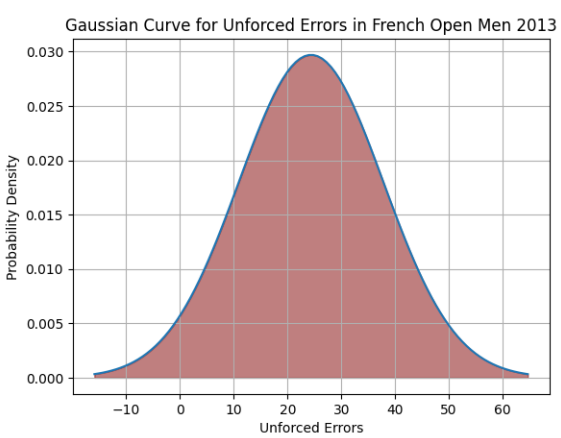

Figure 3.

in the match comes out to be 35 by using the formula (Fig. 3 shows the gaussian distribution) FORMULA OF GAUSSIAN DISTRIBUTION:

$$\frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/2\sigma^2}$$

FORMULA OF MEAN OF A FUNCTION:

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i.$$

The probability of player 1, given that he commits an average number of unforced errors, is calculated using the formula,

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

The probability when calculated using this formula comes out to be 0.5.

C. Given the joint probability distribution between ACE.1 and ACE.2, what is the probability that both players will have at least one ace in the match in Wimbledon-women-2013 dataset?

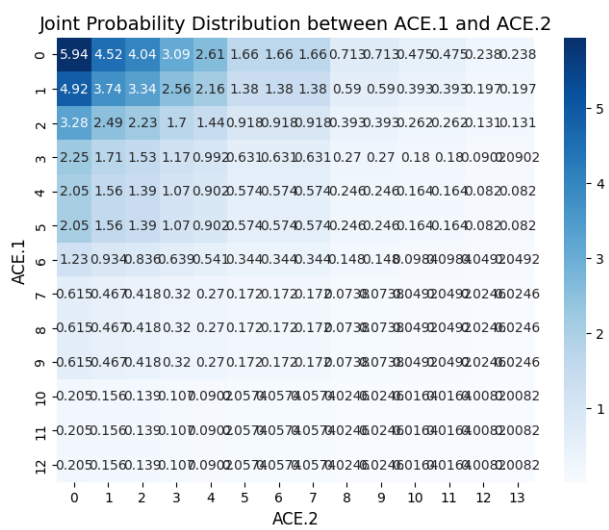

Figure 4.

Fig.4 shows the joint probability distribution of ACE.1 and ACE. 2, we can see that the probability of both players having at least one ace in the match is relatively high, with the darkest blue region being in the top left corner of the plot.

The Probability that both players will have at least one ace in the match: 0.42, hence this confirms the above Fig. 4

FORMULA OF GAUSSIAN DISTRIBUTION:

$$\frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/2\sigma^2}$$

D. What is the probability that Player 1 will win given that they have won more net points than Player 2 in AusOpen-men-2013 dataset?

The probability that Player 1 will win, given that they have won more net points than Player 2 in the AusOpen-men-2013 dataset is 0.46. Fig. 5 shows the distribution of

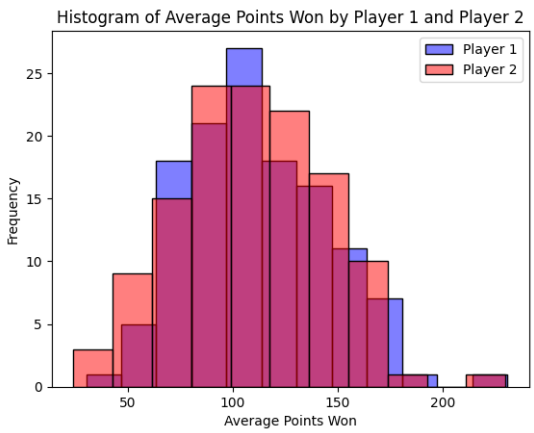

Figure 5.

total points earned by player 1 and 2. The average points earned by player 1 is approximately 110, and for player two is 100, more variance in the net points is seen in the distribution for player 2.

E. What is the distribution of first-serve percentages for both players? What is the correlation between FSP and FSW value in US Open-men-2013?

The mean first-serve percentages for player 1 comes out to be 58.65, while the standard deviation is 7.11.(Fig. 6)

The mean first-serve percentages for player 2 comes out to be 58.92, while the standard deviation is 7.40. (Fig. 6)

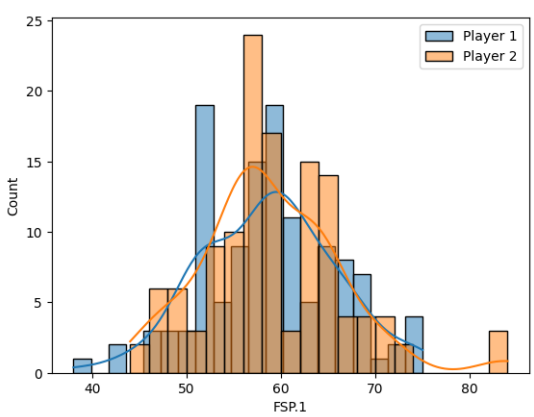

Figure 6.

The correlation factor lies between -1 and 1, where a positive value indicates a positive correlation and a negative value indicates a negative correlation. A value of 0 indicates no correlation. The closer the absolute value of the coefficient is to 1, the stronger the correlation.

The correlation coefficient between First Serve Percentage and First Serve Winnings is calculated using the formula:

$$r = \frac{\sum_{i=1}^{n}(x_i-\bar{x})(y_i-\bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i-\bar{x})^2}\sqrt{\sum_{i=1}^{n}(y_i-\bar{y})^2}}$$

Where n is the number of observations x_i is the value of the ith observation of the first variable, y_i is the value of the ith observation of the second variable, $\bar{x}$ is the mean of the first variable, $\bar{y}$ is the mean of the second variable. Correlation factor when evaluated using this formula comes out to be 0.2645949591876309.

F. Is there a correlation between a player's net points attempted and their number of winners earned in FrenchOpen-women-2013 dataset?

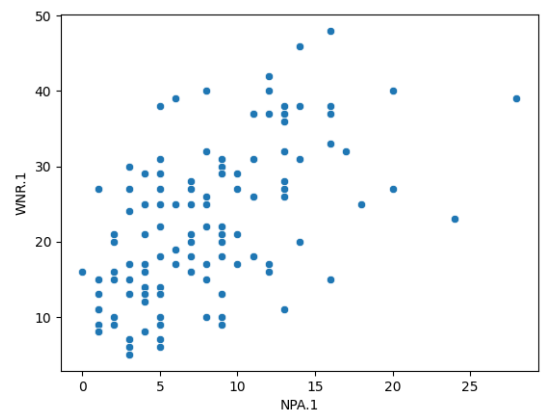

Figure 7.

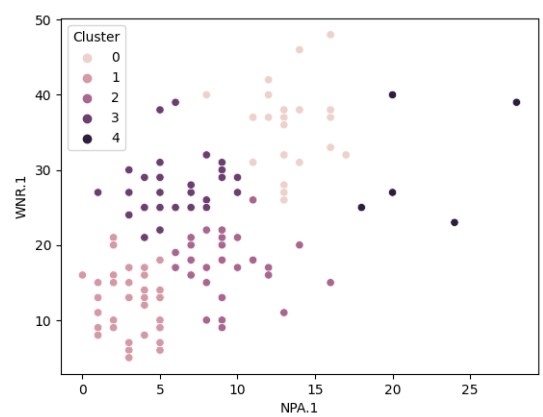

Figure 8.

From Fig.7 we can see the scatter plot of player's net points attempted Vs their number of winners earned, Fig.

9 is the kmeans clustering of the same. The optimum number of clusters is five, out of which three clusters can be seen at a lower range of x-axis. The majority of the players lie in these three clusters. Based on this information, it may be possible to conclude that there is a relationship between the number of net points attempted and the number of winners earned, and that this relationship can be used to group players into different categories based on their playing style. However, further analysis would be needed to confirm these conclusions.

The correlation factor lies between -1 and 1, where a positive value indicates a positive correlation and a negative value indicates a negative correlation. A value of 0 indicates no correlation. The closer the absolute value of the coefficient is to 1, the stronger the correlation.

The correlation coefficient between player's net points attempted and their number of winners earned in FrenchOpen-women-2013 dataset is calculated using the formula:

$$r = \frac{\sum_{i=1}^{n}(x_i-\bar{x})(y_i-\bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i-\bar{x})^2}\sqrt{\sum_{i=1}^{n}(y_i-\bar{y})^2}}$$

The correlation factor when evaluated comes out to be 0.588556116326309.

G. What is the probability of a player having a second serve percentage given that he/she has a high number of total points in USOpen-women-2013 dataset?

The probability of a player having a second serve percentage given that he/she has a high number of total points in USOpen-women-2013 dataset can be evaluated using the following formula:

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

where $P(A|B)$ is the conditional probability of A given B, $P(AB)$ is the probability of the intersection of A and B, and P(B) is the probability of B. The probability comes out to be 0.84375.

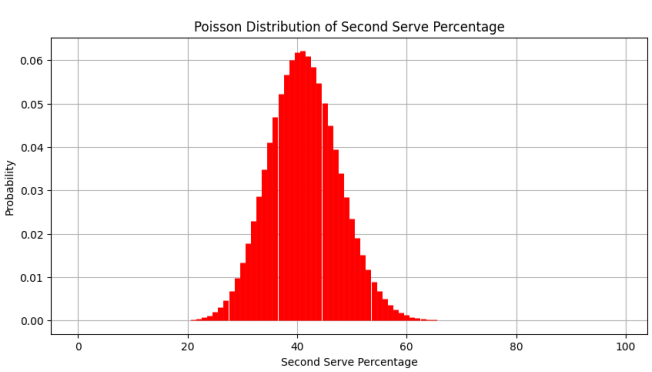

Figure 9.

Fig.9 shows the Poisson distribution of the second serve percentage in the given dataset. The Poisson distribution is a probability distribution that can be used to model the number of times an event occurs in a fixed interval of time or space. In this case, the distribution shows the probability of a player having a certain second serve percentage. The peak of the distribution is around 60 percent, indicating that this is the most common second serve percentage in the dataset. As the percentage increases or decreases from this value, the probability of it occurring decreases.

FORMULA OF POISSON DISTRIBUTION:

$$P(x) = \frac{e^{-\lambda}\lambda^x}{x!}$$

H. What is the probability that a player wins a match given that they won more points in the first set than their opponent in Aus-Women-2013 dataset?

The probability of a player winning a match given that they won more points in first set in USOpen-women-2013 dataset can be evaluated using the following formula:

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

where $P(A|B)$ is the conditional probability of A given B, $P(AB)$ is the probability of the intersection of A and B, and P(B) is the probability of B.

The probability comes out to be zero. Fig.10 shows

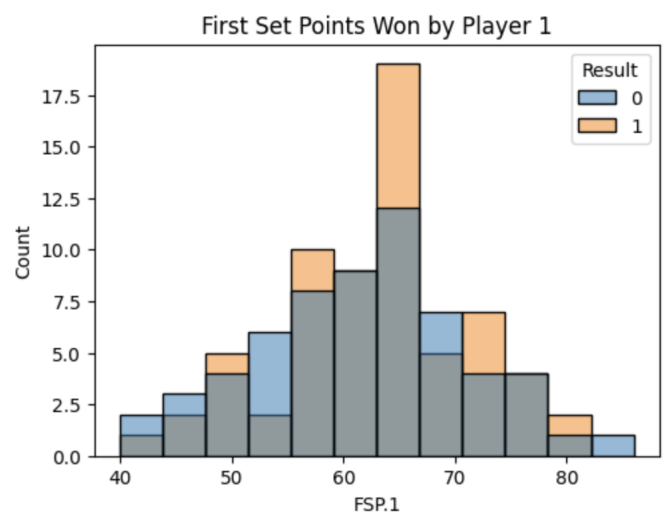

Figure 10.

histograms that show the distribution of the first set points won by Player 1. Fig.11 shows histograms that show the distribution of the first set points won by Player 2. Fig. 12 shows a black dashed line that represents the boundary

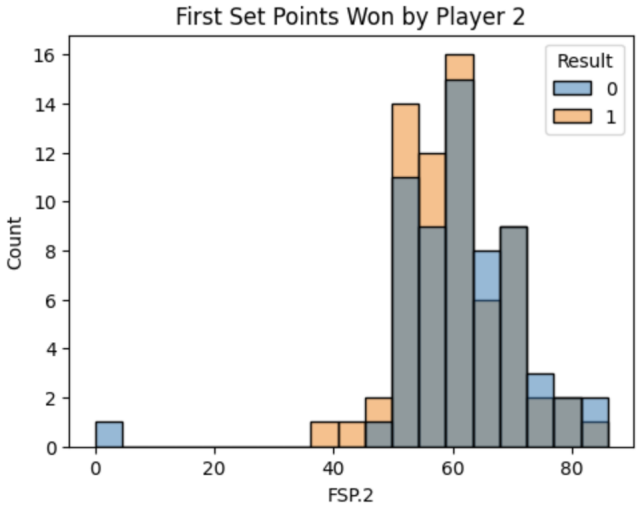

Figure 11.

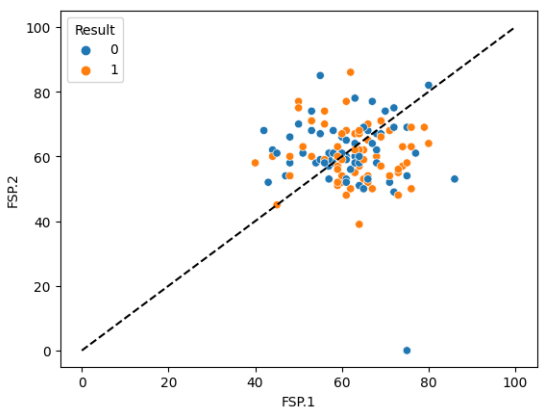

Figure 12.

where Player 1 wins more points in the first set than their opponent.

V. Summary of the Observations

A. The R-squared scores indicate that the number of winner points has a stronger correlation with first-serve percentage compared to unforced errors.

B. If Player 1 commits 10 unforced errors in the match, the probability that they will still win the match in French Open men 2013 comes out to be 0.5. The gaussian distribution of the frequency of unforced errors, give us the means as 35.

C. he joint probability distribution between ACE.1 and ACE.2 shows us that the probability of both players winning at least one ACE in the match is relatively higher. The probability that both players will have at least one ace in the match in Wimbledon-women-2013 dataset comes out to be 0.42.

D. The probability that Player 1 will win given that they have won more net points than Player 2 in AusOpen- men-2013 dataset comes out to be 0.42.

E. The distribution of first-serve percentages for Player 1 gives us the mean as 58.65 and the standard deviation as 7.11, and that for player 2 is 58.92 and 7.40, respectively. The correlation between FSP value and FSW value in US Open-men-2013 comes out to be 0.26

F. There a correlation between a player's net points attempted and their number of winners earned in FrenchOpen-women-2013 dataset comes out to be 0.59.

G. The probability of a player having a second serve percentage given that she has a high number of total winner points in the US Open-women-2013 dataset come out to be 0.84. The poisson distribution of Second Serve Percentage shows us that the most common serve percentage is around 60 percent.

H. he probability that a player wins a match given that they won more points in the first set than their opponent in Aus-Women-2013 dataset comes out to be zero.

VI. Unanswerable Questions

In all the hypotheses questions, the conclusions are drawn on the basis of the data given in a particular dataset. What is we try to draw some observations by analysing all datasets as a pool? Will there be a significant change in the above predictions or probabilities?

Acknowledgment

Professor Shanmuganathan Raman, Department Of Computer Science And Engineering and Electrical Engineering, Indian Institute Of Technology Gandhinagar.

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